ECON 770 Recitation 7

Sept 25, 2020

University of North Carolina at Chapel Hill

Contents

- Conditional Expectation
- Inequality

Conditional Expectation

Definition

Suppose that random variables (X, Y) have support $S_X \times S_Y$, then conditional expectation $\mathbb{E}[Y|X]$ is defined as

- If Y is discrete,

$$\mathbb{E}[Y|X] = \sum_{y \in S_Y} y P(Y = y|X)$$

- If Y is continuous,

$$\mathbb{E}[Y|X] = \int_{S_Y} y f_{Y|X}(y|X) dy$$

Note that $\mathbb{E}[Y|X]$ is a function of X (e.g. $\mathbb{E}[Y|X = x]$), so it is a random variable.

(**Exercise**: Suppose $f(x, y) = e^{-y}$ for $0 < x < y < \infty$, you can show that $\mathbb{E}[Y|X = x] = 1 + x$.)

Conditional Expectation Properties

Theorem

Suppose we have random variables (X, Y, Z) s.t. conditional expectations exist, then

- For any $a, b, c \in \mathbb{R}$,

$$\mathbb{E}[a + bY + cZ|X] = a + b\mathbb{E}[Y|X] + c\mathbb{E}[Z|X]$$

- $\mathbb{E}[g(X)|X] = g(X)$
- $\mathbb{E}[Yg(X)|X] = g(X)\mathbb{E}[Y|X]$

In other words, given enough information, we can factor out random variable from the expectation.

Law of Iterated Expectation (LIE)

Theorem

Given existed expectations, average conditional mean = mean, i.e.

$$\mathbb{E}\left[\mathbb{E}\left[Y|X\right]\right] = \mathbb{E}\left[Y\right]$$

We also have more general LIE. Suppose X, Y, Z are R.V.s, then

Theorem*

$$\mathbb{E}\left[Y|X\right] = \mathbb{E}\left[\mathbb{E}\left[Y|X,Z\right]|X\right] = \mathbb{E}\left[\mathbb{E}\left[Y|X\right]|X,Z\right]$$

If we further define filtration (information set), then we have

Theorem*

If filtration $\mathcal{G} \subset \mathcal{H}$, then $\mathbb{E}\left[Y|\mathcal{G}\right] = \mathbb{E}\left[\mathbb{E}\left[Y|\mathcal{H}\right]|\mathcal{G}\right] = \mathbb{E}\left[\mathbb{E}\left[Y|\mathcal{G}\right]|\mathcal{H}\right]$

Conditional Variance

Given that

- $Var(Y) = \mathbb{E}\left[Y^2\right] - (\mathbb{E}[Y])^2$
- $Var(Y|X) = \mathbb{E}\left[Y^2|X\right] - (\mathbb{E}[Y|X])^2$

we can show that

$$Var(Y) = \mathbb{E}[Var(Y|X)] + Var(\mathbb{E}[Y|X])$$

(as an exercise of applying LIE)

Conditional Variance (Solution)

By definition and LIE,

$$
\begin{aligned}
\mathbb{E}[\mathrm{Var}(Y \mid X)] &= \mathbb{E}\left\{\mathbb{E}\left[Y^2 \mid X\right] - [\mathbb{E}(Y \mid X)]^2\right\} \\
&= \mathbb{E}\left\{\mathbb{E}\left[Y^2 \mid X\right]\right\} - \mathbb{E}\left\{[\mathbb{E}(Y \mid X)]^2\right\} \\
&= \mathbb{E}\left(Y^2\right) - \mathbb{E}\{\mathbb{E}(Y \mid X)]^2\} \\
\mathrm{Var}(\mathbb{E}(Y \mid X)) &= \mathbb{E}\left\{[\mathbb{E}(Y \mid X)]^2\right\} - \{\mathbb{E}[\mathbb{E}(Y \mid X)]\}^2 \\
&= \mathbb{E}\left\{[\mathbb{E}(Y \mid X)]^2\right\} - \{\mathbb{E}[Y]\}^2,
\end{aligned}
$$

therefore, ...

Covariance and Correlation

Definition

Given R.V.s X and Y with finite expected value, then

- Covariance

$$Cov(X,Y) = \mathbb{E}\left[(X-\mu_X)(Y-\mu_Y)\right] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$$

- Correlation

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

where $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$, $\sigma_X = \sqrt{Var(X)}$ and $\sigma_Y = \sqrt{Var(Y)}$.

Covariance and Correlation (Illustration)

Covariance and correlation only depict the linear dependence

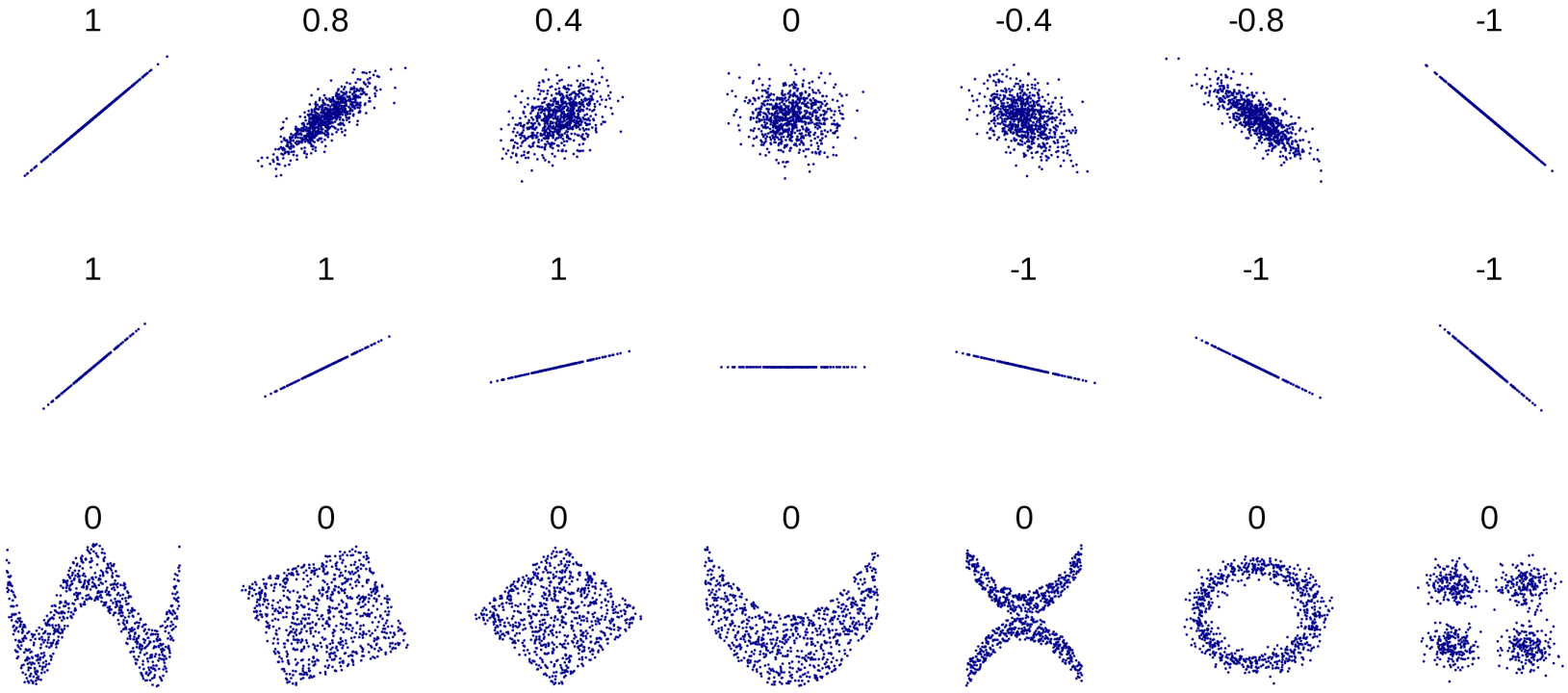

Figure: Several sets of (x, y) points, with the Pearson correlation coefficient of x and y for each set. (From Wikipedia)

Covariance and Correlation Properties

Properties

- Symmetry: $Cov(X, Y) = Cov(Y, X)$ and $Corr(X, Y) = Corr(Y, X)$
- $Cov(X, Y) > 0 \quad \Leftrightarrow \quad Corr(X, Y) > 0$
- $|Corr(X, Y)| \leq 1$ (By Cauchy-Schwartz Inequality)
- Independence $\Rightarrow Corr = 0$

Recall that

$$Var(a + bX + cY) = Var(bX + cY) = b^2 Var(X) + c^2 Var(Y) + 2bcCov(X, Y)$$

If X and Y are independent, then $Cov(X, Y) = 0$ and

$$Var(a + bX + cY) = b^2 Var(X) + c^2 Var(Y)$$

Exogeneity

Suppose that we are estimating $Y = \alpha + \beta X + \varepsilon$ where $\varepsilon \sim N(0,1)$ and $\mathbb{E}[X\varepsilon] = 0$. Then, we can make full use of moment condition to estimate α and β.

Since $\varepsilon = Y - \alpha - \beta X$, then at first we have

$$0 = \mathbb{E}[\varepsilon] = \mathbb{E}\left[Y - \alpha - \beta X\right] = \mathbb{E}[Y] - \alpha - \beta\mathbb{E}[X]$$

i.e. $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$. Second,

$$\begin{aligned} 0 &= \mathbb{E}[X\varepsilon] \\ &= \mathbb{E}\left[X(Y - \alpha - \beta X)\right] \\ &= \mathbb{E}\left[XY - \alpha X - \beta X^2\right] \\ &= \mathbb{E}[XY] - \alpha\mathbb{E}[X] - \beta\mathbb{E}\left[X^2\right] \end{aligned}$$

Exogeneity (Contd)

Combine $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$ and $\mathbb{E}[XY] - \alpha\mathbb{E}[X] - \beta\mathbb{E}\left[X^2\right] = 0$, we have

$$\mathbb{E}[XY] - \mathbb{E}[Y]\mathbb{E}[X] + \beta(\mathbb{E}[X])^2 - \beta\mathbb{E}\left[X^2\right] = 0$$

which means

$$\beta = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\mathbb{E}[X^2] - (\mathbb{E}[X])^2} = \frac{Cov(X,Y)}{Var(X)}$$

By using $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$, we can solve

$$\alpha = \mathbb{E}[Y] - \frac{Cov(X,Y)}{Var(X)}\mathbb{E}[X] = \quad \ldots$$

Instrument Variables*

Again, consider $Y = \alpha + \beta X + \varepsilon$ and $\varepsilon \sim N(0, 1)$, but now

$$\mathbb{E}[X\varepsilon] \neq 0, \ \mathbb{E}[W\varepsilon] = 0$$

where W is another random variable. It can be show that generally

$$\beta \neq \frac{Cov(X, Y)}{Var(X)}$$

For simplicity, let $\alpha = 0$, then how can we get β based on (X, W, Y)?

Correlation

Since $Cov(X, \varepsilon) = 0 \Rightarrow Corr(X, Y) = \dfrac{bVar(X)}{\sqrt{Var(X)}\sqrt{b^2 Var(X) + Var(\epsilon)}}$,

then

$$Corr(X, Y) = \frac{\frac{b}{|b|}}{\sqrt{1 + \frac{Var(\epsilon)}{b^2 Var(X)}}} \text{ if } b \neq 0 \text{ and } Var(X) \neq 0$$

- If $Var(\epsilon) \to 0$, ...
- If $Var(X) \to \infty$, ...
- If $|b| \to \infty$, ...

Inequalities

Assuming all the following moments exist,

Cauchy-Schwartz Inequality

$$|\mathbb{E}[XY]| \leq (\mathbb{E}\left[X^2\right])^{\frac{1}{2}}(\mathbb{E}\left[Y^2\right])^{\frac{1}{2}}$$

Hölder's Inequality

For any $p, q \in [1, \infty)$ s.t. $\frac{1}{p} + \frac{1}{q} = 1$,

$$|\mathbb{E}[XY]| \leq (\mathbb{E}[|X|^p])^{\frac{1}{p}}(\mathbb{E}[|Y|^q])^{\frac{1}{q}}$$

Minkowski's Inequality

For any $p \geq 1$,

$$(\mathbb{E}[|X+Y|^p]^{\frac{1}{p}}) \leq (\mathbb{E}[|X|^p])^{\frac{1}{p}} + (\mathbb{E}[|Y|^p])^{\frac{1}{p}}$$